Lavannya Pradeep Rathod

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SUMMARY

Recent graduate skilled in Python and SQL with hands-on experience in machine learning, data analysis, and AI-based applications. Proficient in predictive modeling, data preprocessing, feature engineering, and exploratory data analysis (EDA). Experienced in building data-driven solutions and creating visualizations with Pandas, Scikit-learn, and Power BI. Strong interest in applying analytical skills to solve real-world business problems.

SKILLS

- **Programming:** Python, SQL, HTML
- **Data Analysis:** Pandas, NumPy, Data Cleaning, Feature Engineering, Exploratory Data Analysis (EDA)
- **Machine Learning:** Scikit-learn, Logistic Regression, Random Forest, Model Evaluation, Classification
- **Data Visualization:** Power BI, Tableau, Matplotlib, Microsoft Excel
- **Deep Learning:** LSTM, Time-Series Analysis, Neural Networks
- **Web & Deployment:** Streamlit, REST APIs, Model Deployment
- **Tools:** Git, MySQL, Jupyter Notebook, Visual Studio Code

PROJECTS

1. Heart Disease Detection System

Tools: Python, Pandas, Scikit-learn, Machine Learning, Streamlit

- Built a machine learning model using Logistic Regression and Random Forest on a dataset with 300+ records and 13+ clinical features
- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) to enhance model performance
- Assessed model effectiveness, achieving approximately 85% accuracy with strong precision and recall metrics
- Deployed a Streamlit-based interface enabling real-time predictions derived from user-provided health parameters
- Utilized machine learning techniques for predictive modeling and classification tasks

2. AI Blog Generator

Tools: Python, Streamlit, OpenAI API, HTML/CSS

- Engineered an AI-powered web application using Streamlit and OpenAI APIs capable of generating complete blog articles in under 5 seconds
- Constructed backend logic in Python for prompt processing and dynamic content generation and Incorporated export functionality supporting three formats including TXT, HTML, and Markdown
- Designed an interactive user interface ensuring seamless content generation and real-time input handling

3. E-commerce Customer Behavior Analysis

Tools: Python, Pandas, NumPy, Matplotlib, Power BI

- Executed exploratory data analysis (EDA) on an e-commerce dataset with 1000+ transaction records
- Applied data cleaning and preprocessing techniques across multiple features for structured evaluation
- Produced 5+ visualizations and dashboards with Power BI and Matplotlib to highlight key business insights
- Derived actionable insights to support business decision-making and improve customer engagement strategies

4. Automated Data Analytics Platform

Tools: Python, Pandas, NumPy, Plotly, NLP

- Created a Python-based data analytics system capable of processing Excel and CSV datasets with multiple input features
- Executed data cleaning, statistical analysis, and visualization workflows and Incorporated NLP techniques for analyzing text-based data within datasets
- Generated interactive reports with 4+ visual insights to support data-driven decision-making

EXPERIENCE

Computer Literacy Volunteer: Government College

- Assisted in teaching basic computer skills including MS Office, internet usage, and typing to 30+ students across multiple sessions
- Mentored learners in understanding fundamental programming concepts and delivered hands-on guidance during 10+ lab sessions
- Troubleshot 20+ technical issues related to software and system usage, ensuring smooth lab operations
- Strengthened digital literacy and confidence among students through practical learning activities

EDUCATION

The Oxford College of Engineering, Bangalore

Bachelor of Engineering in Electronics and Communication Engineering — Dec 2021 – May 2025

CGPA: 7.9/10

CERTIFICATIONS

- SQL for Beginners – MySQL and Database Design (Udemy)
- Microsoft Excel – From Beginner to Expert
- Microsoft Power BI – Advanced Data Transformation
- SQL(Advanced) - HackerRank